

Your local movie theatre has put in a lot of work to make their customer's movie going experience great. They would like to take a random sample of their recent customers and email out a survey asking about their experience.

1) For the following situations, identify the bias and explain how they would introduce bias into the results.

a) You send out an email to the first 10 customers that buy tickets to a movie.

b) Of the 10 movies that are playing in a single day, you randomly select one movie and email the survey to everyone who attended that movie.

c) You set up a sign as the patrons exit the theatre and tell patrons to email support@movies.com if they would like to share their movie going experience.

d) You have the manager of the movie theatre waiting outside the movie theatre and interviewing every 10th person who walks out about their movie going experience.

You obtain a list of the following data in a month at the movie theater: Order ID, Date, Customer Name, Customer Email, Movie Title, Show Time, Number of Tickets Purchased.

2) You would like to email out a survey to a sample of customers. For the following sampling methods, describe how you would use the information above to select a sample.

a) Simple Random Sample

b) Stratified Random Sample

c) Cluster Random Sample

d) Systematic Random Sample

e) A Census